

Engineered Features and Business Logic Justification

This document details the features engineered in the data cleaning process and their strategic purpose for the customer value analysis.

1. Value_Tier

Definition: Categorizes customers based on Purchase Amount (USD) . - High : > 75th percentile (\$68) - Medium : 25th to 75th percentile (\$39 - \$68) - Low : < 25th percentile (\$39)

Business Reason: Allows the marketing team to segment the customer base and prioritize resources towards the most valuable customers rather than treating all customers equally.

2. Retention_Segment

Definition: Categorizes customers based on Previous Purchases . - Loyal : ≥ 8 previous purchases (Top 50% of the customer base) - Potential Loyalist : 4 to 7 previous purchases (25th to 50th percentile) - At Risk : < 4 previous purchases (Bottom 25% of the customer base)

Business Reason: Since timestamps are not available in this dataset, this acts as a proxy for engagement. It helps the retention team identify which customers need re-engagement campaigns and which are solid brand advocates.

3. Promo_Segment

Definition: Categorizes customers based on their usage of Discount Applied and Promo Code Used . - Organic : Never used a discount or promo code - Discount Dependent : Always used a discount and a promo code - Moderate Promo : Mixed usage (e.g., used a discount but no promo code, or vice versa)

Business Reason: Critical for understanding promotional dependency. It helps leadership identify if revenue is being driven by brand loyalty or artificially inflated by discounting strategies that erode profit margins.

4. CLV_Proxy (Customer Lifetime Value Proxy)

Definition: $\text{Purchase Amount (USD)} * (\text{Previous Purchases} + 1)$ **Business Reason:**

Provides a simplified metric to estimate the total historical revenue generated by a customer. This allows us to compare the lifetime profitability of different segments (e.g., organic vs. discount-dependent).

5. Loyalty_1 vs Loyalty_2

The problem required comparing two definitions of loyalty:

- **Loyalty_1 (Behavioral):** Purely based on $\text{Previous Purchases}$.
- **Loyalty_2 (Profitable):** $\text{Previous Purchases} * \text{Review Rating} * (1 \text{ for Organic, } 0.5 \text{ for Moderate, } -0.5 \text{ for Discount Dependent})$.

Business Reason: Loyalty_1 failed to account for margin erosion. A customer buying 50 times with a discount is less valuable than one buying 30 times organically. Loyalty_2 penalizes discount dependency and aligns better with revenue goals, making it the recommended loyalty metric.